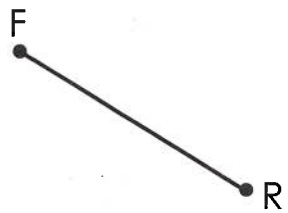
GM5aZv1

Directions: Answer each question.

1. Choose the name that could be given to this line segment.



- $\overline{F}$
- $\overline{FF}$
- $\overline{FR}$

2. What do the following symbols refer to?

$\overleftrightarrow{RS}$, $\overleftrightarrow{LM}$, $\overleftrightarrow{YZ}$

- line segment
- lines
- endpoints

3. a. line b. line segment c. ray

_____ Straight path that goes on forever in one direction.

_____ Straight path that goes on forever in two directions.

_____ Part of a line that has 2 endpoints.

GM5aZ7.8

Level 5a

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Name: Ojas

Date: / /

Time: ____ : ____ TO ____ : ____

Week 26

Z1

Level Review: Addition and Subtraction of Fractions

Ivan spends three days preparing for his final exam. The table shows the days and times he studied.

Day	Time (in hours)
Monday	$\frac{3}{4}$ of an hour
Tuesday	$\frac{2}{3}$ of an hour
Wednesday	$1\frac{3}{12}$ hours

How much total time did he spend studying for the exam?

$\frac{\cancel{3}9}{\cancel{4}12} + \frac{\cancel{2}8}{\cancel{3}12} + 1\frac{3}{12} = 1\frac{20}{12} = 2\frac{8}{12} = 2\frac{2}{3}$

a. $2\frac{7}{12}$ hours

b. $1\frac{5}{12}$ hours

c. $1\frac{3}{4}$ hours

d. $2\frac{2}{3}$ hours

GM5aZ1.1

Directions: Solve. Make sure your answer is a proper fraction in lowest terms.

1

$3\frac{3}{10} + 5\frac{\cancel{1}5}{\cancel{2}10} = 8\frac{8}{10} = 8\frac{4}{5}$

$8\frac{4}{5}$

2

$4\frac{\cancel{2}14}{\cancel{3}21} - 4\frac{\cancel{2}6}{\cancel{7}21} = \frac{8}{21}$

$\frac{8}{21}$

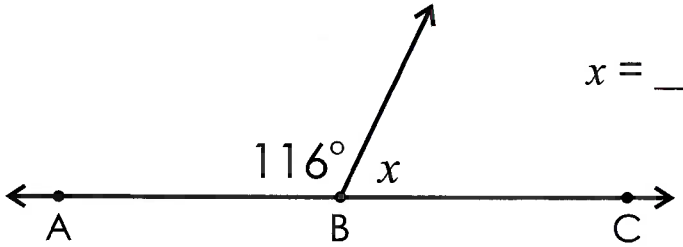
Directions: Complete the chart.

Sentence	Number Sentence	Answer in Simplest Terms
Sum of eight-fifths and two-ninth	$\frac{\cancel{8}72}{\cancel{5}45} + \frac{\cancel{2}10}{\cancel{9}45}$	$\frac{82}{45} = 1\frac{37}{45}$
sum of eight forty-eighths and one-fourth	$\frac{8}{48} + \frac{\cancel{1}12}{\cancel{4}48}$	$\frac{20}{48} = \frac{5}{12}$

GM5aZ1.2

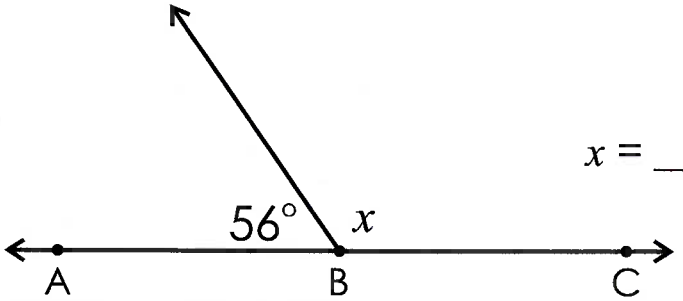
Directions: Find the missing angle measurement.

(1)



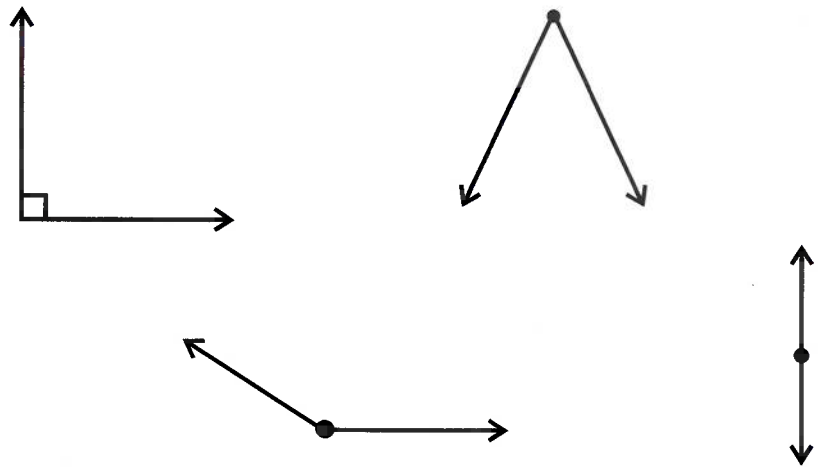
$x =$ _____

(2)



$x =$ _____

Directions: Circle all the obtuse angles.



GM5aZ7.7

Directions: Determine if each statement is true or false.

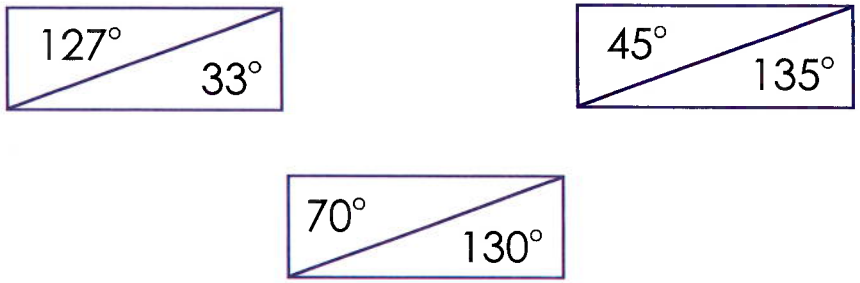
1. All supplementary angles must be adjacent.

a. True b. False

2. A supplementary angle can also be a complementary angle.

a. True b. False

Directions: Circle the number pairs that would create supplementary angles.



GM5aZ7.6

Directions: Solve and then determine if the statement is true or false.

$$\frac{\cancel{9}\,27}{\cancel{4}\,12} + \frac{\cancel{2}\,4}{\cancel{6}\,12} > 1\frac{\cancel{7}\,5}{\cancel{3}\,15} + 2\frac{\cancel{2}\,6}{\cancel{5}\,15}$$

$$\frac{31}{12} = 2\frac{7}{12} \not> 3\frac{11}{15}$$

☐ True ☒ False

Directions: Choose from the given numbers to fill in the missing boxes.

2, 3, 10

(1) $6\frac{3}{4} - 2\frac{1}{\boxed{10}} = 4\frac{13}{20}$

(2) $1\frac{1}{\boxed{2}} + 4\frac{2}{3} = 6\frac{1}{6}$

GM5aZ1.3

Lester made a fruit salad with $3\frac{5}{6}$ pounds of melon and $1\frac{5}{6}$ pounds of berries. How many pounds of fruit did Lester use in all?

$3\frac{5}{6} + 1\frac{5}{6} = 4\frac{10}{6} = 5\frac{4}{6} = 5\frac{2}{3}$

$5\frac{2}{3}$

Directions: Solve. Make sure your answer is a proper fraction in lowest terms.

1. $7\frac{4}{5} - 3\frac{\cancel{2}1}{\cancel{10}5} = 4\frac{3}{5}$

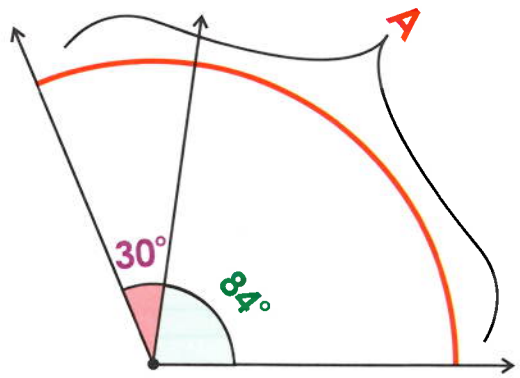
$4\frac{3}{5}$

2. $1\frac{\cancel{1}3}{\cancel{2}6} + 4\frac{\cancel{2}4}{\cancel{3}6} = 5\frac{7}{6} = 6\frac{1}{6}$

$6\frac{1}{6}$

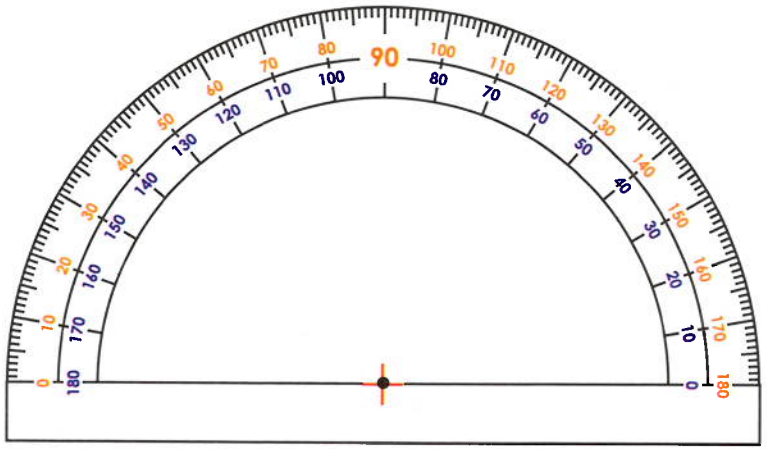
GM5aZ1.4

Directions: Find the measurement of angle A.



Directions: Use the protractor to draw an angle with the given measurement.

35°

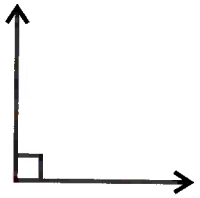


GM5aZ7.5

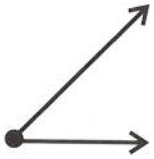
Directions: Classify each angle as obtuse, acute, right or straight.

(1)

(2)

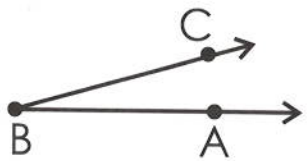


(3)



(4)

Directions: Which option best represents ∠ ABC?



a. 71°

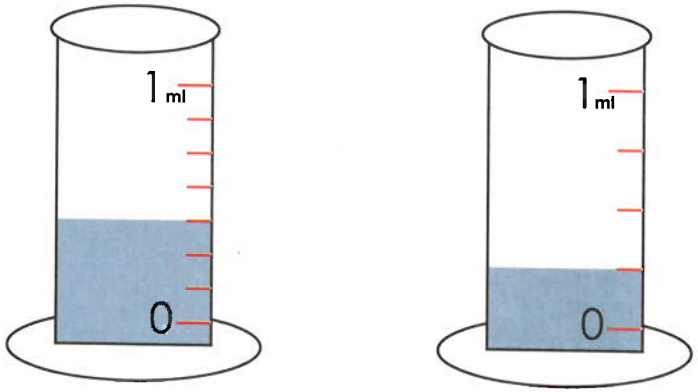
b. 52°

c. 21°

d. 79°

GM5aZ7.4

In science class you need to combine the liquid from two different graduated cylinders.



Solution A

Solution B

What is the total amount of fluid in the solution?

$\frac{3}{7}\frac{12}{28} + \frac{1}{4}\frac{7}{28} = \frac{19}{28}$ ml

a. $\frac{12}{28}$ ml

b. $\frac{19}{28}$ ml

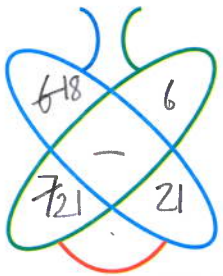
c. $\frac{7}{28}$ ml

d. $\frac{5}{28}$ ml

GM5aZ1.5

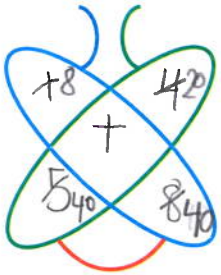
Directions: Make sure your fractions are in lowest terms before using the butterfly method to solve.

$$\frac{6}{7} - \frac{6}{21} = \boxed{\frac{12}{21} = \frac{4}{7}}$$



$$= \frac{12}{21} = \frac{4}{7}$$

$$\frac{1}{5} + \frac{4}{8} = \boxed{\frac{28}{40} = \frac{7}{10}}$$



$$= \frac{28}{40} = \frac{7}{10}$$

Directions: Complete the chart.

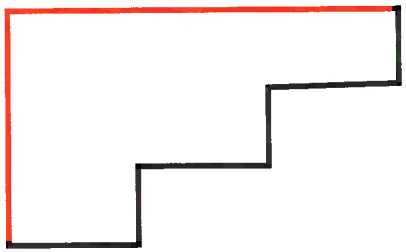
Sentence	Number Sentence	Answer in Simplest Form
eight-fourths plus twelve-ninths	$\frac{\cancel{8}\,72}{\cancel{4}\,36} + \frac{\cancel{12}\,48}{\cancel{9}\,36}$	$\frac{120}{36} = 3\frac{18}{36} = 3\frac{1}{2}$
thirteen-sevenths minus nine-twenty one	$\frac{\cancel{13}\,39}{\cancel{7}\,21} - \frac{9}{21}$	$\frac{30}{21} = 1\frac{9}{21} = 1\frac{3}{7}$

Directions: Fill in the blanks.

A _____ is a straight path that extends forever in _____ directions.

Draw an example.

Directions: Determine what type of lines the red lines represent.



a. Perpendicular

b. Parallel

c. Intersecting

Directions: Fill in the chart.

Vocabulary	Definition	Picture
Line Segment		
Ray		

Directions: Which is **not** a true statement?

a. A line has one endpoint.

b. A ray has one endpoint.

c. A line segment has two endpoints.

GM5aZ7.2

Directions: Solve and determine if each sum is greater than or less than $\frac{1}{2}$.

a. $\frac{2}{9} + \frac{\cancel{1}\,3}{\cancel{3}\,9} = \frac{5}{9}$

☒ greater than
☐ less than

b. $\frac{2}{16} + \frac{\cancel{1}\,4}{\cancel{4}\,16} = \frac{6}{16} = \frac{3}{8}$

☐ greater than
☒ less than

c. $\frac{\cancel{4}\,1}{\cancel{20}\,5} + \frac{3}{5} = \frac{4}{5}$

☒ greater than
☐ less than

Directions: Choose from the given numbers to fill in the missing boxes.

12, 9, 3

(1) $\frac{3}{\boxed{12}} + \frac{9}{24} = \frac{15}{24}$

(2) $\frac{2}{2} - \frac{1}{\boxed{3}} = \frac{2}{3}$

GM5aZ1.7

Level 5a

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Name

Date / /

Time ____ : ____ TO ____ : ____

Week 26

Z7

Kathy filled cups of paint for a craft project. She filled $\frac{3}{8}$ of a cup with pink paint. After she finished the project, there was $\frac{1}{4}$ cup of paint left over. How much paint did she use?

$\frac{3}{8} - \frac{\cancel{4}2}{\cancel{4}8} = \frac{1}{8}$

Directions: Solve and write your answer in simplest form.

1

$\frac{5}{8} + \frac{\cancel{8}4}{\cancel{16}8} = \frac{9}{8} = 1\frac{1}{8}$

$1\frac{1}{8}$

2

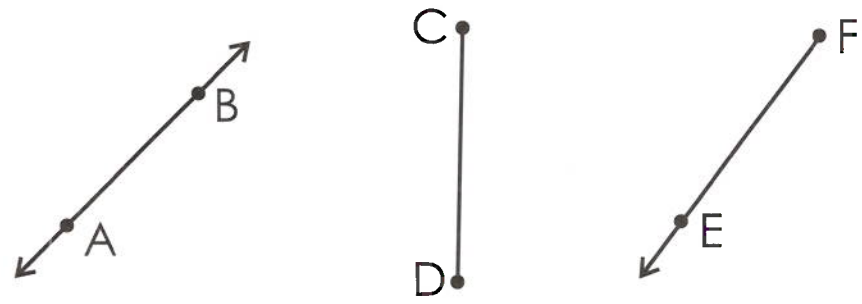
$\frac{\cancel{13}26}{\cancel{5}10} - \frac{21}{10} = \frac{5}{10} = \frac{1}{2}$

$\frac{1}{2}$

GM5aZ1.8

Level Review: Lines and Angles

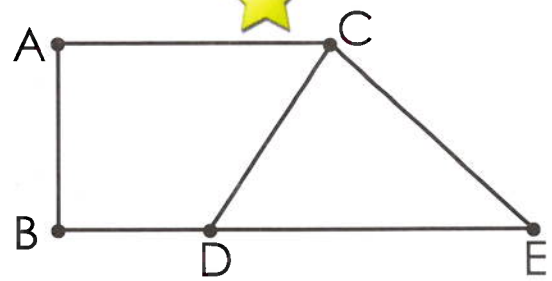
Directions: Label the pictures as a line(L), line segment(LS) or ray(R).



_____ _____ _____

Directions: How many line segments are there in this picture?

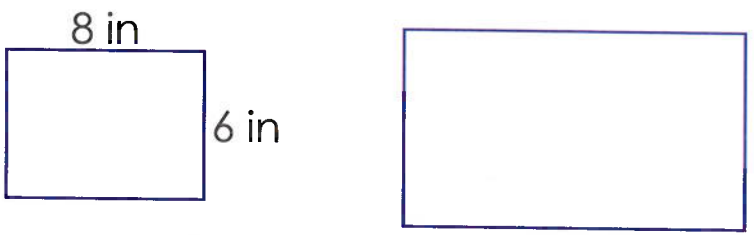
Challenge



a. 5 b. 6 c. 7

GM5aZ7.1

Directions: Determine the lengths of the sides of the larger rectangle if the scale factor is 7.



Directions: Determine whether each statement is true or false.

1. _____ A circle and an oval can be congruent.

2. _____ A scale factor is used in congruent and similar shapes.

3. _____ Angles are congruent in similar shapes.

4. _____ Tick marks are used to show equal measurements.

Level Review: Multiplication and Division of Fractions

The chart shows different driving times from Niagara Falls to various cities.

	Route	Hours of Travel
1	NYC to Niagara Falls	$6\frac{1}{2}$ hours
2	Niagara Falls to Toronto	$\frac{1}{5}$ of the time from NYC to Niagara
3	Buffalo, NY to Niagara Falls	$\frac{1}{2}$ of the time of number 2

(1) How long does it take to get from Niagara to Toronto?

$6\frac{1}{2} \times \frac{1}{5} = \frac{13}{2} \times \frac{1}{5} = \frac{13}{10} = 1\frac{3}{10}$ hours

(2) How long would it take to get from Toronto to Buffalo, NY?

Nathan won $567 in an art competition. Of that he gave $\frac{1}{3}$ to his mom. Of the remaining money he kept $\frac{1}{2}$ and gave the rest to charity. How much money did Nathan give away?

① $567 \times \frac{1}{3} = \frac{567}{3} = \$189 = \text{mom}$

② $567 - 189$

$\$378$ left

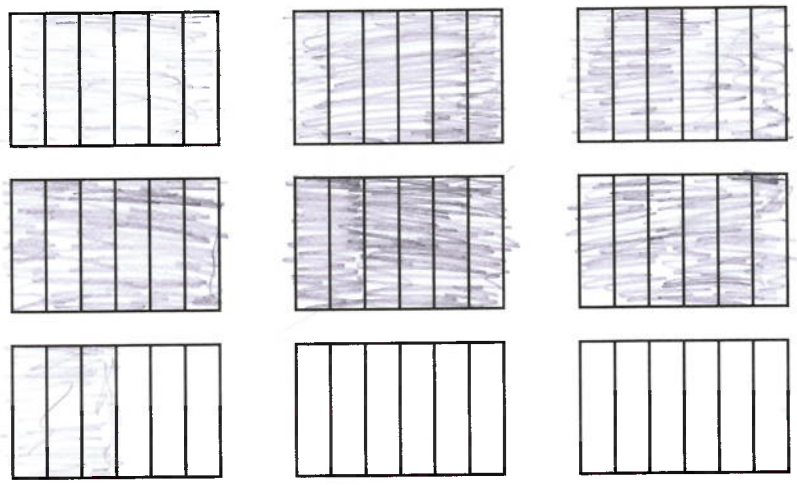
③ $\frac{1}{2} \times \frac{378}{1} = \frac{378}{2} = \$189 = \text{gave to charity}$

④ $189 + 189$

$\$378$ Nathan gave away

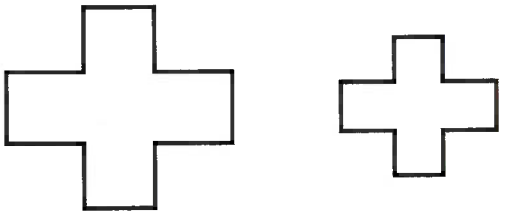
Directions: Represent $2\frac{1}{6} \times 3$ by shading to solve.

$2\frac{1}{6} \times 3 = 6\frac{3}{6} = 6\frac{1}{2}$



GM5aZ2.2

Directions: Choose the correct answer.



a. The shapes are only congruent.

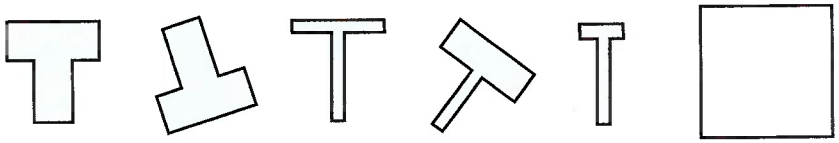
b. The shapes are only similar.

c. The shapes are both similar and congruent.

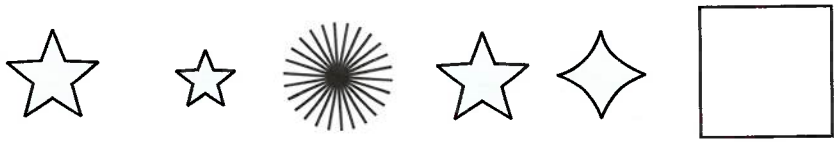
d. The shapes are neither similar nor congruent.

Directions: Circle the congruent shapes in each row. Then draw a similar shape in the box.

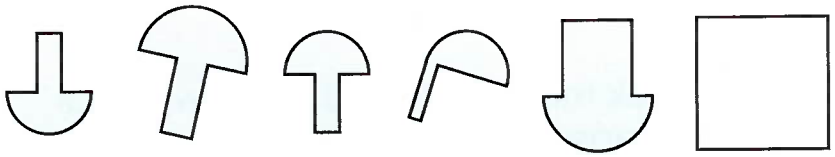
1.



2.

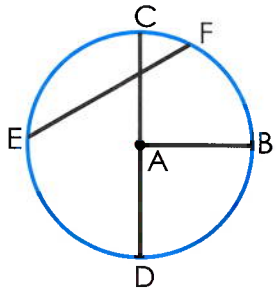


3.



GM5aZ6.7

Directions: Identify the chord in the diagram below.



a. $\overline{EF}$

b. $\overline{AB}$

c. $\overline{AD}$

d. $\overline{CD}$

You bought your brother a basketball as a gift. If you wanted to wrap a ribbon around it, which question should you ask?

a. What is the diameter?

b. What is the radius?

c. What is the circumference?

GM5aZ6.6

Directions: Estimate to determine the best answer.

$3\frac{2}{3} \times 5 =$

$4 \times 5 = 20$

a. 9

b. 18

c. 20

d. 15

Directions: Solve and insert <, > or =.

$2\frac{2}{9} \times \frac{1}{6}$ $\boxed{>}$ $\frac{3}{7} \times \frac{1}{4}$

$\frac{20}{9} \times \frac{1}{6} = \frac{20}{45} = \frac{4}{9} > \frac{3}{28}$

GM5aZ2.3

Directions: Solve using the repeated addition model.

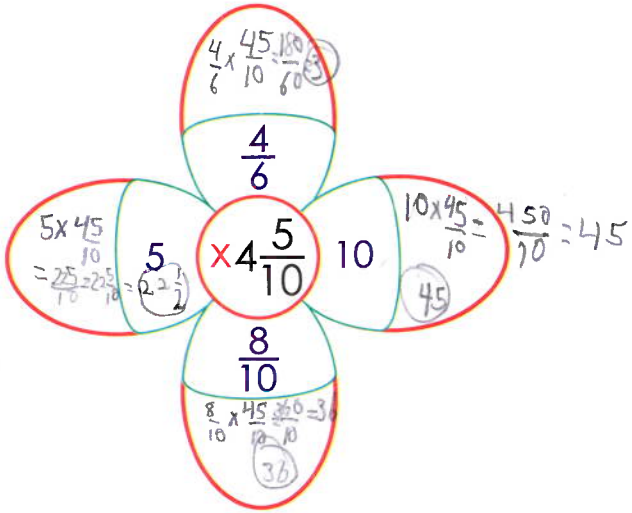
$3 \times 4\frac{1}{4}$ = 3 groups of $4\frac{1}{4}$

$3 \times 4\frac{1}{4} = 3 \times \frac{17}{4} = \frac{51}{4} = 12\frac{3}{4}$

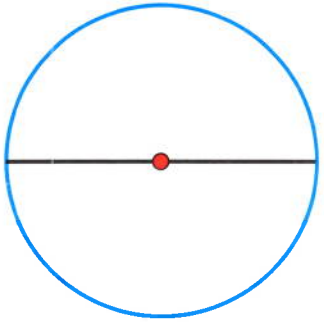
$4 \times 2\frac{3}{5}$ = 4 groups of $2\frac{3}{5}$

$4 \times 2\frac{3}{5} = 4 \times \frac{13}{5} = \frac{52}{5} = 10\frac{2}{5}$

Directions: Multiply each number by the center number.



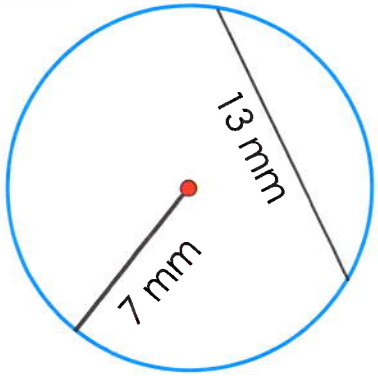
Directions: Calculate the diameter of the circle.



$r = 2\frac{3}{4}$

Directions: Fill in the blanks.

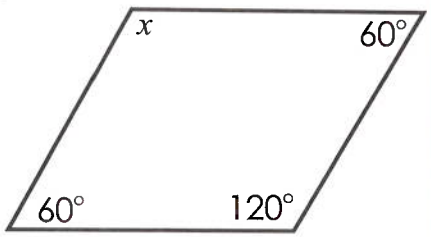
The radius of the circle is _____.



The diameter of the circle is _____.

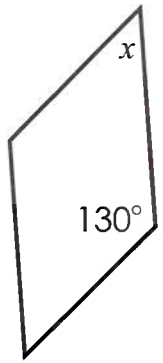
Directions: The sum of the interior angles of a rhombus is 360°. Find the missing angle.

1



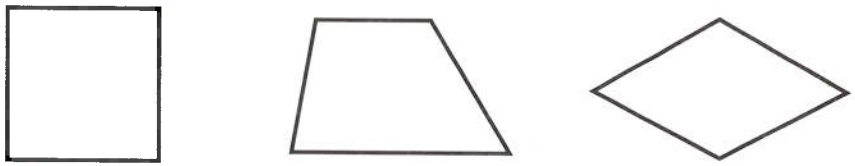
$x =$ _____

2



$x =$ _____

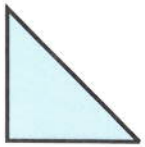
Directions: Circle all the trapezoids. Place an **x** on all the rhombuses.



GM5aZ6.4

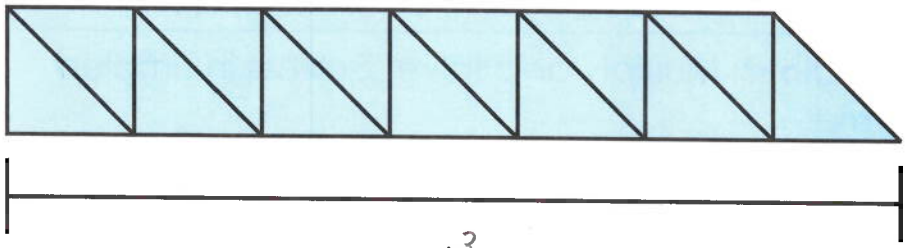
Directions: Determine the value.

Given



$= \frac{3}{4}$

then,



? = $9\frac{3}{4}$

$13 \times \frac{3}{4} = \frac{39}{4} = 9\frac{3}{4}$

GM5aZ2.5

There is $\frac{1}{2}$ of a pitcher of orange juice in the fridge. At breakfast your family drank $\frac{2}{3}$ of this amount. How much orange juice did your family drink at breakfast?

$\frac{1}{2} \times \frac{2}{3} = \frac{2}{6} = \frac{1}{3}$

$\frac{1}{3}$

Directions: Multiply and solve. Express in simplest form.

1. $\frac{6}{9} \times \frac{2}{3} = \frac{12}{27} = \frac{4}{9}$

2. $\frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$

3. $\frac{2}{6} \times \frac{1}{5} = \frac{2}{30} = \frac{1}{15}$

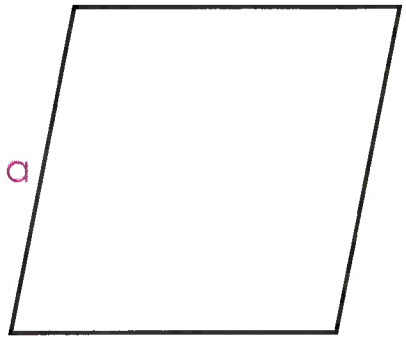
Directions: Which of the following is equal to the perimeter of the rhombus?

a. 4a

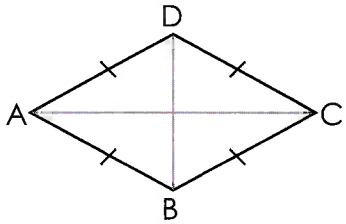
b. 4(a + a)

c. 4(a + 4)

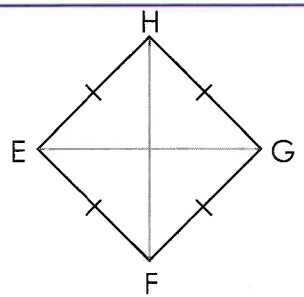
d. 2a + 4a



Directions: To find the area of a rhombus, multiply the diagonals and divide by 2. Find the area.



$\overline{AC}$ = 10 cm
$\overline{BD}$ = 13 cm

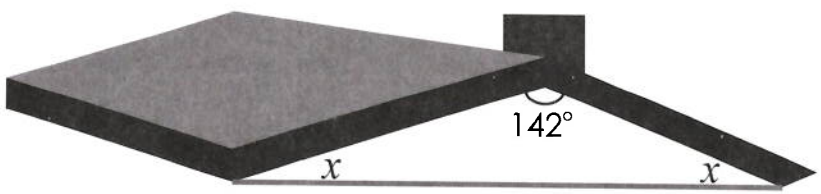


$\overline{EG}$ = 6 cm
$\overline{HF}$ = 8 cm

Directions: Sketch the triangle listed using hash marks.

Isosceles	Scalene

Directions: Find the value of x. Then classify the triangle as isosceles, scalene or equilateral.



GM5aZ6.2

Directions: Create an area model to solve.

1. $\frac{1}{2} \times \frac{3}{8} = \boxed{\frac{3}{16}}$

2. $\frac{3}{4} \times \frac{2}{5} = \boxed{\frac{6}{20} = \frac{3}{10}}$

Directions: Rewrite as a multiplication problem and solve.

1. What is $\frac{1}{3}$ of $\frac{2}{7}$?

$\frac{1}{3} \times \frac{2}{7} = \frac{2}{21}$

2. What is $\frac{4}{3}$ of 5?

$\frac{4}{3} \times \frac{5}{1} = \frac{20}{3} = 6\frac{2}{3}$

GM5aZ2.7

The product of $\frac{1}{2}$ and $\frac{3}{8}$ is $\frac{3}{16}$.

$\frac{1}{2} \times \frac{3}{8} = \frac{3}{16}$

a. $\frac{3}{10}$

b. $\frac{2}{6}$

c. $\frac{3}{16}$

d. $\frac{4}{10}$

A maid can clean 6 bedrooms in one hour. How many bedrooms can she clean in $\frac{2}{3}$ of an hour?

$6 \times \frac{2}{3} = \frac{12}{3} = 4$

4 bedrooms

GM5aZ2.8

Level 5a

bestbrains LEARNING CENTERS

Name

Date / /

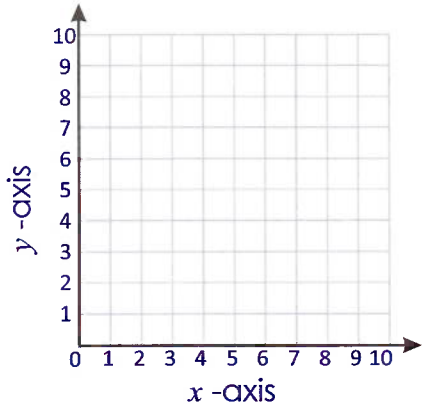
Time ____ : ____ TO ____ : ____

Week 26

Z6

Level Review: Shapes

Directions: Graph the coordinates.



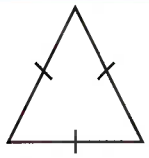
a. (2, 3)

b. (5, 8)

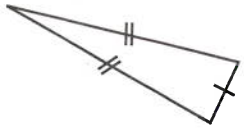
c. (2, 10)

What type of triangle was made? _____

Directions: Label equilateral triangles with an E, isosceles with an I, and scalene with an S.

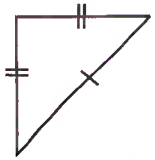


_____



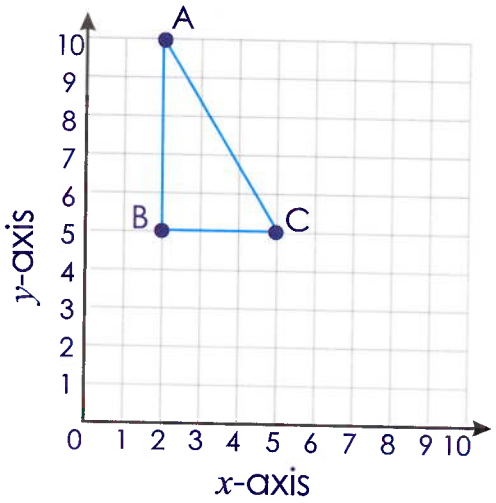
_____

_____



_____

GM5aZ6.1



Translate

1 Unit right

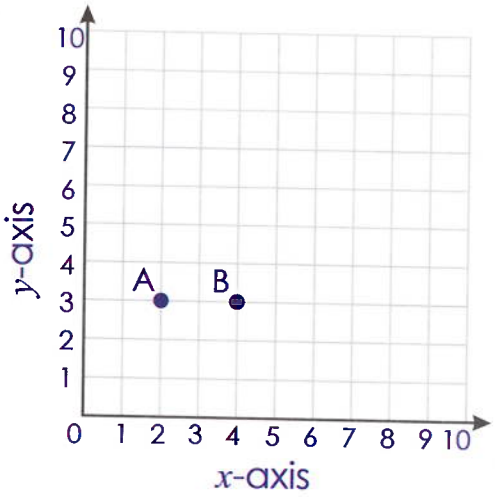
2 Units down

New Coordinates

A. _____ , _____

B. _____ , _____

C. _____ , _____



1) Which axis is the point translated on?

a. x-axis b. y-axis

2) Is it moving left/right or up/down?

a. left/right b. up/down

3) Original Coordinate: _____

New Coordinate: _____

GM5aZ5.8

Level 5a

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Name

Date / /

Time ____ : ____ TO ____ : ____

Week 26

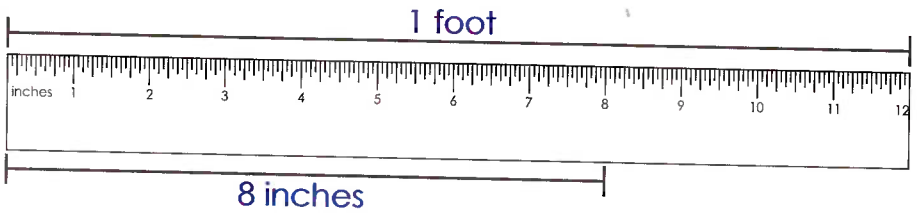
Z3

Level Review: Ratios and Proportions

Directions: Ratios can be written in three different forms. Complete the chart.

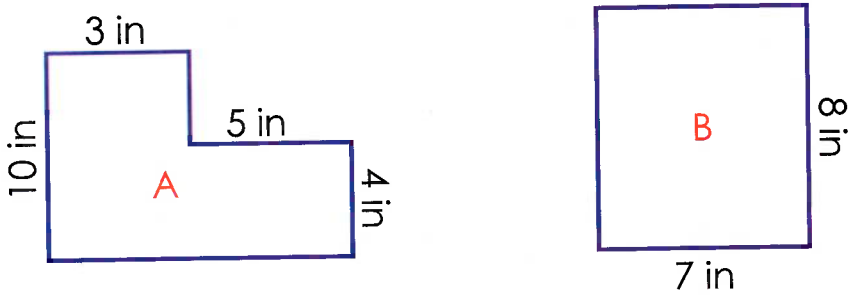
Symbol	Written	Fraction
2 : 5	2 to 5	$\frac{2}{5}$
	9 to 10	
1 : 3		
		$\frac{6}{7}$

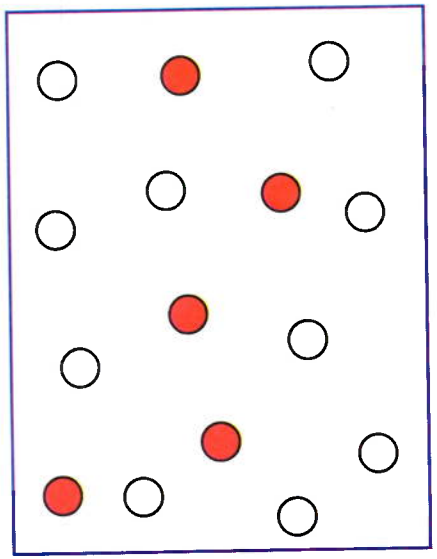
Directions: Write the ratio 1 foot to 8 inches as a fraction in simplest form.



GM5aZ3.1

Directions: Write a ratio comparing the area of shape A with the area of shape B.





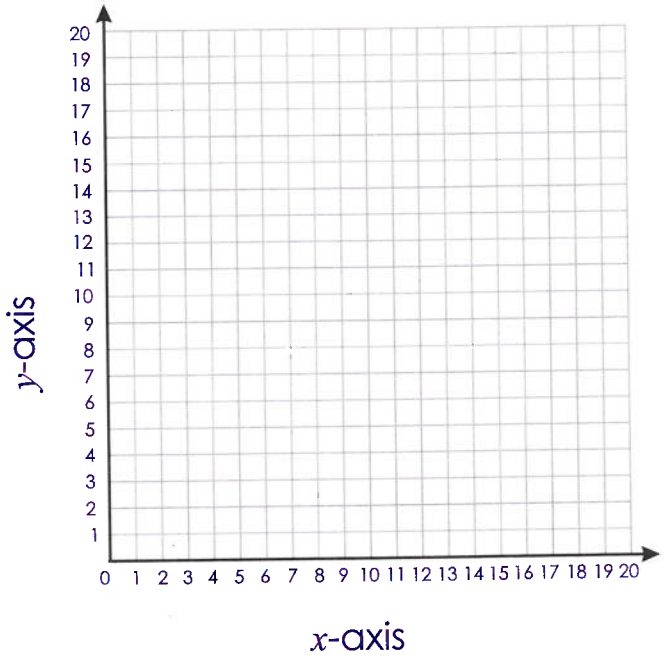
List the ratio of all the ○ to ●.

_____

List the ratio of all the ○ to ●+○.

_____

GM5aZ3.2



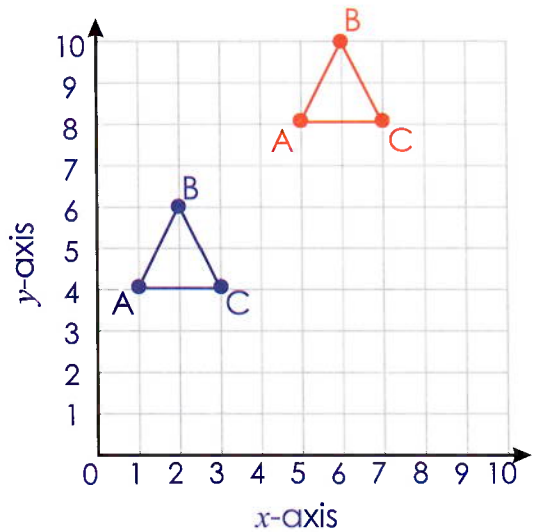
Plot the following points. Then plot the translated points.

Ordered Pair	Translation	Translated Point
(7, 7)	up 2 right 4	
(16, 12)	up 1 right 3	
(8, 0)	up 5 right 2	
(13, 19)	down 3 right 1	

GM5aZ5.7

Directions: Write a rule to describe the transformation. The original shape is in red.

Rule:



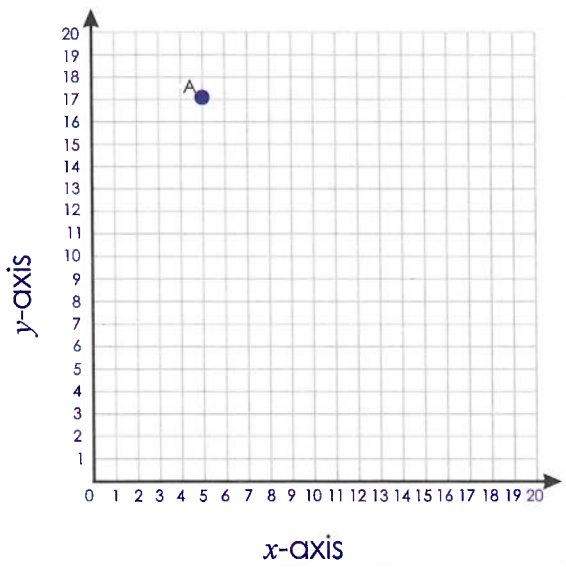
Directions: Determine the point A (5, 17) after the translation. Then plot the translated coordinates.

B $(x - 2, y - 4)$

_____ , _____

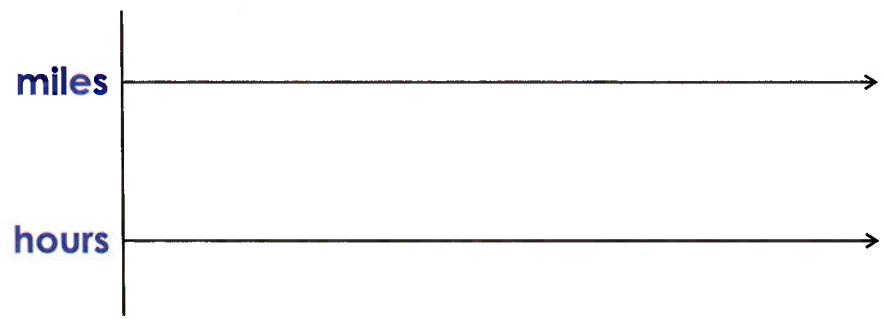
C $(x + 4, y - 9)$

_____ , _____

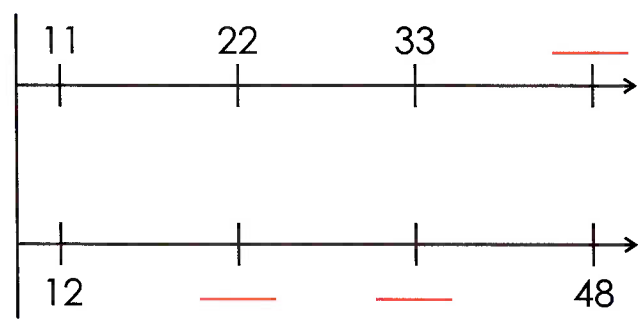


GM5aZ5.6

Sam walks 6 miles in 2 hours. How long will it take him to walk 42 miles?

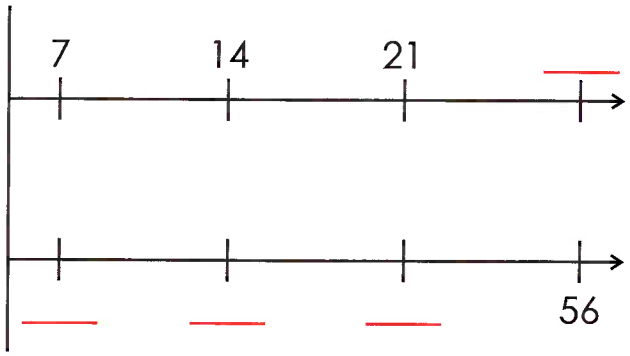


Directions: Complete the double number line.

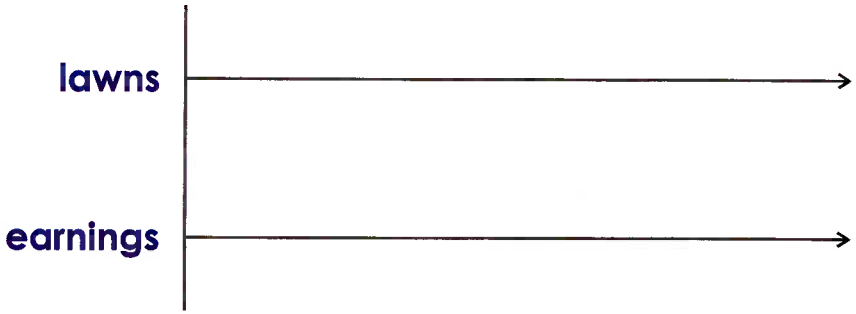


GM5aZ3.3

Directions: Complete the double number line.

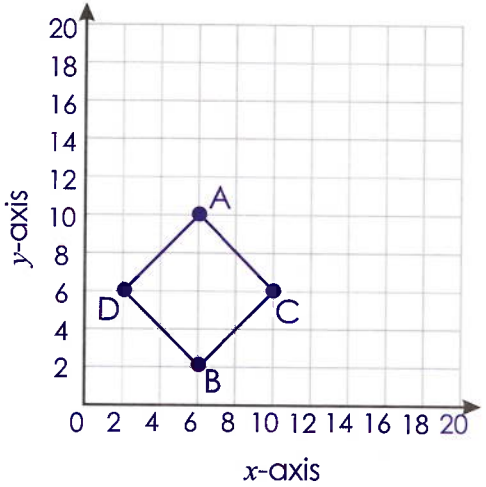


Jerome earns $35 for mowing 7 lawns. How much will he earn to mow 35 lawns?



GM5aZ3.4

Translate 6 units right and 10 units up



A. _____ , _____

B. _____ , _____

C. _____ , _____

D. _____ , _____

Directions: Translate the coordinates 5 units right.

Original	New Coordinates
0, 10	
3, 14	
9, 18	

GM5aZ5.5

Directions: Graph the following points on the coordinate plane.

(1) A(7, 2)

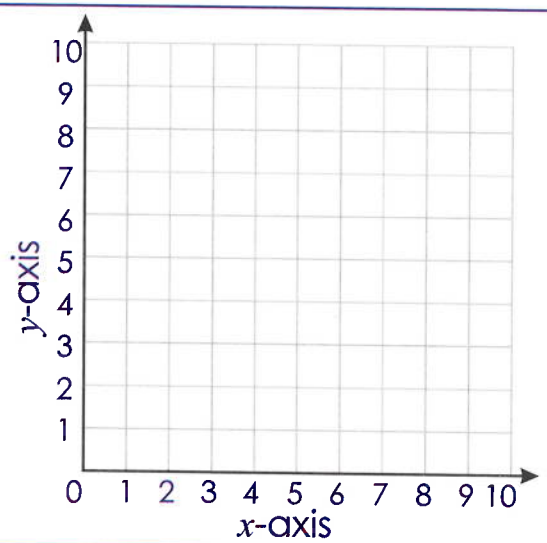
(2) B(5, 6)

(3) C(9, 3)

(4) D(4, 1)

(5) E(0, 8)

(6) F(10, 5)



Calculate the distance point A is from the origin on the x-axis. A = (7, 9)

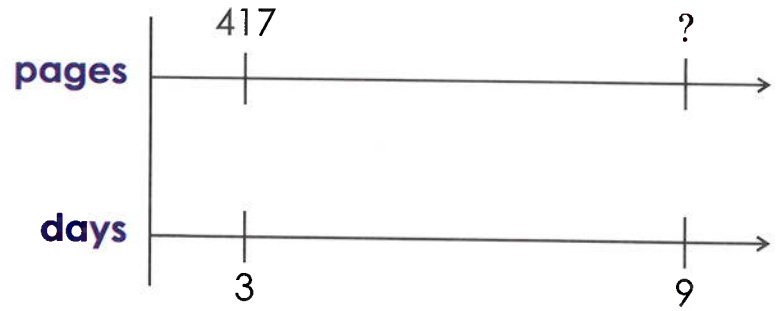
a. 16 units

b. 2 units

c. 7 units

d. 9 units

GM5aZ5.4



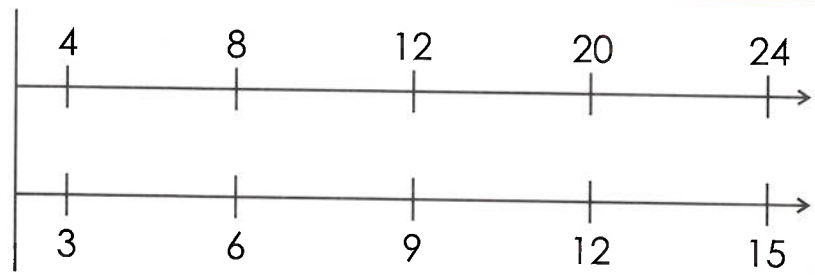
a. 6,352 pages

b. 139 pages

c. 1,251 pages

d. 10,002 pages

Directions: Determine if the double number lines were filled in properly. If not, explain the mistake.



a. correct

b. incorrect

GM5aZ3.5

Directions: Use a tape diagram to answer the question.

For a wall mural, 10 cans of white paint are needed for every 4 cans of black paint. If 42 cans of paint were used, how many cans were black?

Laurel and Maurice shared some hats in the ratio of 3:2. If Laurel had 9 hats, how many did Maurice have?

GM5aZ3.6

Directions: Answer each question.

1. Which coordinates show a point that is 5 units to the right and up 2 units?

   a. (5, 2)
   b. (2, 5)
   c. (2, 0)
   d. (0, 2)

2. Which coordinates show a point that is up 3 units and 7 units to the right?

   a. (0, 7)
   b. (7, 0)
   c. (3, 7)
   d. (7, 3)

3. Which coordinates show a point that is over 4 units and up 1 unit?

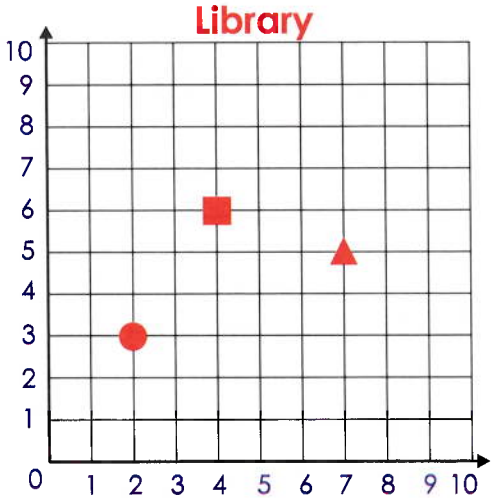
   a. (1, 4)
   b. (4, 0)
   c. (0, 1)
   d. (4, 1)

4. Which two pairs of coordinates show a point that is up 8 units?

   a. (8, 0)
   b. (0, 0)
   c. (8, 8)
   d. (0, 8)

GM5aZ5.3

Directions: Answer the following questions.



▲ Magazines

● Help Desk

■ Children's Books

(1) What are the coordinates for the help desk?

_____

(2) What are the coordinates for the children's books?

_____

(3) What are the coordinates for the magazines?

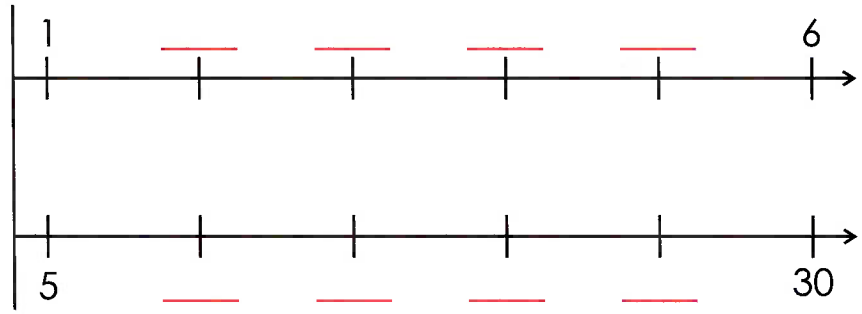
_____

(4) The entrance is located at the origin. Add this to the map. Label it A.

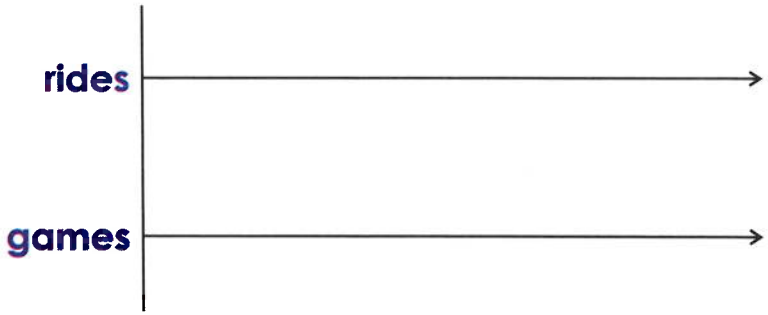
(5) The computer lab is located at (5, 2). Add this to the map. Label it B.

GM5aZ5.2

Directions: Solve using the double number line.



The ratio of rides to games at a carnival is 9:8. Use double number lines to determine how many rides there are if there are 40 games.



GM5aZ3.7

Which table correctly completes the ratio table?

Dogs	Cats
8	5
24	
	20
	25

(a)

D	8	16	24	32	40
C	5	15	20	25	30

(b)

D	8	16	24	32	40
C	5	10	15	20	25

(c)

D	8	16	24	32	40
C	5	15	20	25	30

GM5aZ3.8

Level 5a

bestbrains LEARNING CENTERS

Name

Date / /

Time ____ : ____ TO ____ : ____

Week 26

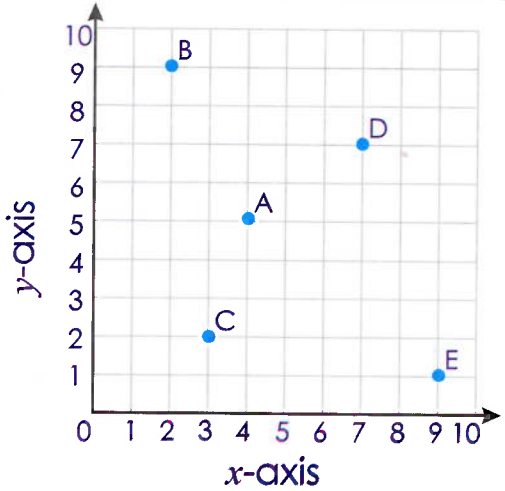
Z5

Level Review: Coordinate Graphing

Directions: Complete the chart.

Ordered pair	x coordinate	y coordinate
(9, 8)		
(5, 2)		
(1, 3)		
(8, 0)		

Directions: Determine what letter is located at each ordered pair.



(1) (3, 2) _____

(2) (4, 5) _____

(3) (7, 7) _____

(4) (2, 9) _____

(5) (9, 1) _____

GM5aZ5.1

Apples come in packages of 10 and bananas in bundles of 12. What is the **least** amount of each you should buy if you want the same amount of each fruit?

a. 12 packages of apples and 10 bundles of apples.

b. 4 packages of apples and 3 bundles of bananas.

c. 5 packages of apples and 4 bundles of bananas

d. 6 packages of apples and 5 bundles of bananas.

Level 5a

bestbrains LEARNING CENTERS

Name

Date / /

Time ____ : ____ TO ____ : ____

Week 26

Z4

Level Review: LCM

Directions: Find the LCM of 3 and 5.

List the multiples of each number.

3: ____, ____, ____, ____, ____, ____, ____, ____, ____, ____

5: ____, ____, ____, ____, ____, ____, ____, ____, ____, ____

The common multiples of 3 and 5 are: _____

The LCM of 3 and 5 is _____

Directions: Insert a <, >, or = symbol.

LCM of 6 and 9 ☐ LCM of 3 and 8

Directions: Find the LCM for each set of numbers.

1. 4 and 5

2. 6 and 10

3. 4 and 12

Directions: Check true or false.

1. The LCM of 7 and 8 is 56.

☐ True ☐ False

2. 18 is the LCM of 18 and 6.

☐ True ☐ False

Directions: Determine the LCM that would be used to solve each problem.

1. $\frac{3}{10} + \frac{1}{6} + \frac{1}{2}$ = New denominator

2. $\frac{1}{6} + \frac{1}{7} + \frac{5}{21}$ = New denominator

Directions: Answer the question.

Why is finding the LCM important when solving questions like the ones above? Give one real world example when you would need to use the LCM.

_____

_____

_____

_____

Directions: List the multiples and find the LCM of the three numbers.

3	
10	
4	

LCM= _____

At your party you have cookies that come in packages of 14, brownies in packs of 7 and jellybeans in packs of 21. What is the least number of each package you should put of to have the same number of each dessert?

GM5aZ4.6

Directions: Which statement is true?

a. 8 is the LCM of 24 and 16.

b. 9 is the LCM of 3 and 9.

c. 12 is the LCM of 12 and 24.

A neon light blinks every 15 seconds. Another light blinks every 9 seconds. When will they blink at the same time?

GM5aZ4.3

Directions: Find the LCM for each set of numbers.

1. 7 and 8 LCM = _____

2. 10 and 40 LCM = _____

3. 3 and 10 LCM = _____

Directions: Fill in the blanks.

1

The LCM of 3 and _____ is 24.

2

_____ is the LCM of 4 and 5.

Directions: Which statement is true?

a. The LCM of 12 and 16 is 4.

b. The LCM of 45 and 30 is 90.

Directions: Use factor trees to find the LCM.

45 60

LCM = _____